Mathematics II

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12/03/2026

You can find the last version of these course materials at

https://mbujosab.github.io/MatematicasII/

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Part VII
Geometrical view of statistics

LECTURE 20: Statistics

Lecture 20

(**Lecture** 20) S-1 Highlights of Lesson 20

Highlights of *Lesson 20*

- Mean
- Standard deviation and variance
- Ordinary Least Squares (OLS)

F1

(**Lecture** 20) S-2 Restriction in statistics and probability

The norm of the constant vector "one" is 1

This does not hold with the dot product in $\mathbb{R}^m$ $(m > 1)$

$$\|\mathbf{1}\|^2 = \langle \mathbf{1} | \mathbf{1} \rangle = \mathbf{1} \cdot \mathbf{1} = \sum_{i=1}^{m} 1 = m.$$

New scalar product in $\mathbb{R}^m$ for statistics

$$\langle \boldsymbol{x} | \boldsymbol{y} \rangle_s = \frac{1}{m}(\boldsymbol{x} \cdot \boldsymbol{y})$$

(so: $\|\mathbf{1}\|^2 = \frac{1}{m}\Big(\mathbf{1} \cdot \mathbf{1}\Big) = 1$)

F2

(**Lecture** 20) S-3 Mean

The mean $\mu_{\boldsymbol{y}}$ is the scalar product of $\boldsymbol{y}$ and 1

$$\mu_{\boldsymbol{y}} = \frac{1}{m}\Big(\mathbf{1} \cdot \boldsymbol{y}\Big), \qquad \text{that is,} \quad \mu_{\boldsymbol{y}} = \frac{1}{m}\sum\nolimits_{i=1}^{m} y_i$$

The mean $\mu_{\boldsymbol{y}}$ is the value by which to multiply 1 to get the orthogonal projection of $\boldsymbol{y}$ onto $\mathcal{L}\big([\mathbf{1};]\big)$

$\overline{\boldsymbol{y}}$: projection of $\boldsymbol{y} \in \mathbb{R}^m$ onto the line $\mathcal{L}\big([\mathbf{1};]\big) \subset \mathbb{R}^m$

$$\boxed{\overline{\boldsymbol{y}} = \mathbf{1}\widehat{a}} \quad \text{and} \quad \boxed{(\boldsymbol{y} - \overline{\boldsymbol{y}}) \perp \mathbf{1} \Rightarrow \tfrac{1}{m}(\boldsymbol{y} - \overline{\boldsymbol{y}}) \cdot \mathbf{1} = 0}$$

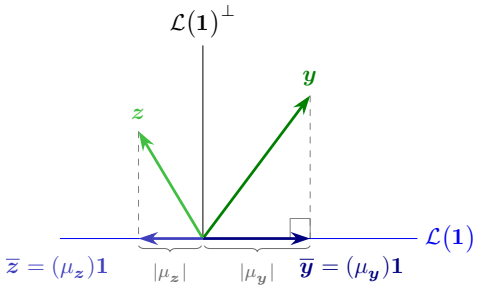
$$\frac{1}{m}(\boldsymbol{y} - \mathbf{1}\widehat{a}) \cdot \mathbf{1} = 0 \Leftrightarrow \frac{1}{m}\Big(\boldsymbol{y} \cdot \mathbf{1}\Big) - \frac{1}{m}\Big(\mathbf{1} \cdot \mathbf{1}\Big)\widehat{a} = 0;$$

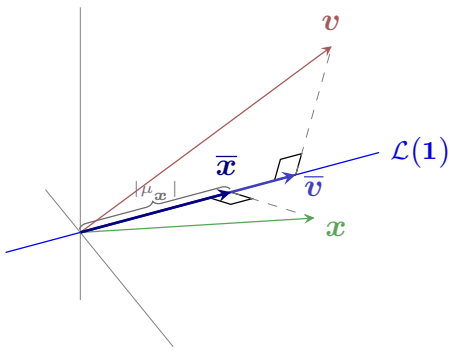
Thus

$$\widehat{a} = \frac{1}{m}\Big(\boldsymbol{y} \cdot \mathbf{1}\Big) = \mu_{\boldsymbol{y}}$$

F3

(**Lecture** 20) **S-4** **Mean**

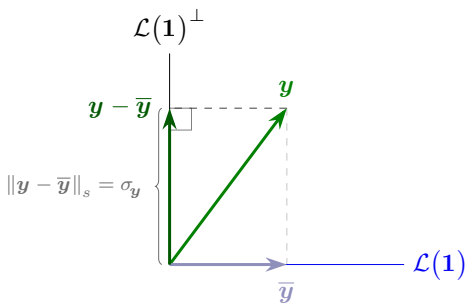


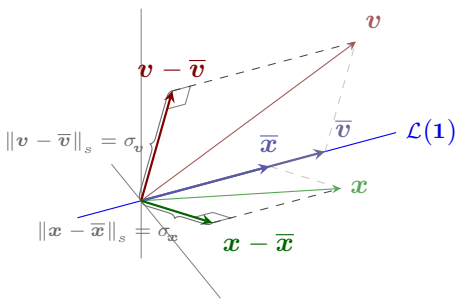


F4

(**Lecture** 20) **S-5** **Standard deviation**

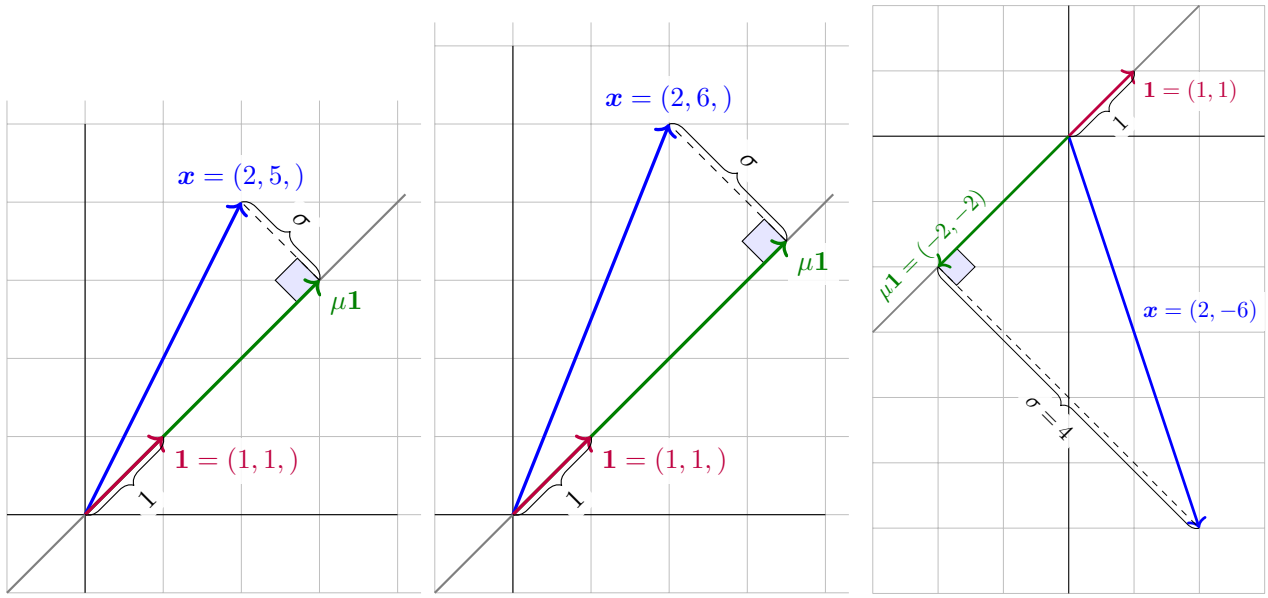
$$\sigma_{\boldsymbol{y}} = \|\boldsymbol{y} - \overline{\boldsymbol{y}}\|.$$





F5

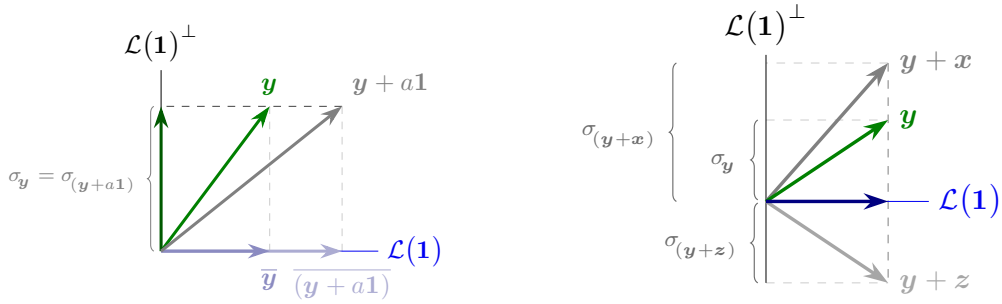
(**Lecture** 20) **S-6** **Mean and standard deviation of two data points**



F6

(**Lecture** 20) **S-7** **Constant Vectors and Zero Mean Vectors**

Adding a constant vector $a\mathbf{1}$ to $\boldsymbol{y}$ does not change the standard deviation.

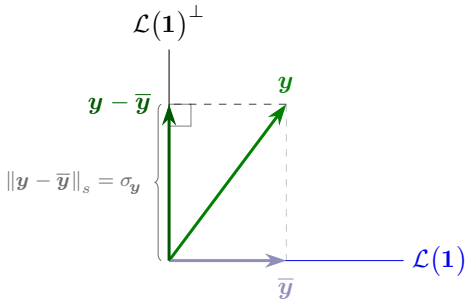


$$\sigma_{\boldsymbol{z}} = 0 \;\Leftrightarrow\; \boldsymbol{z} = a\mathbf{1}; \qquad \mu_{\boldsymbol{z}} = 0 \;\Leftrightarrow\; \boldsymbol{z} \perp \mathbf{1}$$

F7

(**Lecture** 20) **S-8** **Variance and the Pythagorean theorem**

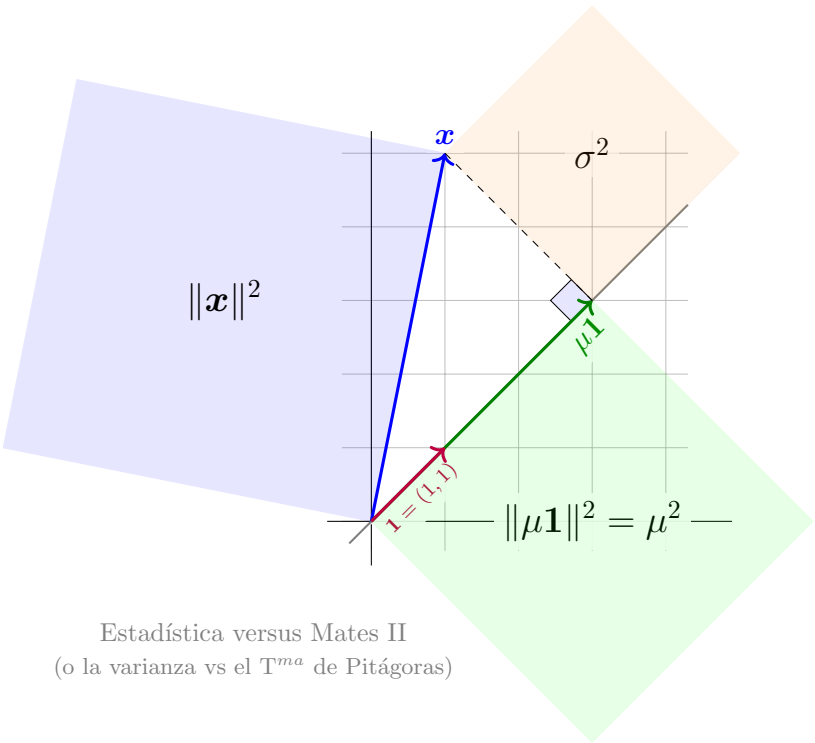
$$\sigma_{\boldsymbol{y}}^2 \;=\; \|\boldsymbol{y}-\overline{\boldsymbol{y}}\|^2 \;=\; \frac{1}{m}(\boldsymbol{y}-\overline{\boldsymbol{y}})\cdot(\boldsymbol{y}-\overline{\boldsymbol{y}}) \;=\; \frac{1}{m}\sum_i (y_i-\mu_{\boldsymbol{y}})^2.$$



$$\sigma_{\boldsymbol{y}}^2 \;=\; \|\boldsymbol{y}-\overline{\boldsymbol{y}}\|^2 \;=\; \|\boldsymbol{y}\|^2-\|\overline{\boldsymbol{y}}\|^2 \;=\; \frac{1}{m}\Big(\boldsymbol{y}\cdot\boldsymbol{y}\Big)-\mu_{\boldsymbol{y}}^2, \;=\; \frac{\sum_i y_i^2}{m}-\mu_{\boldsymbol{y}}^2.$$

F8

(**Lecture** 20) **S-9** **Variance and the Pythagorean theorem (two data points)**

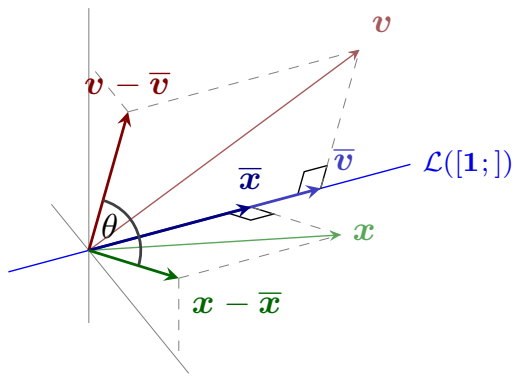


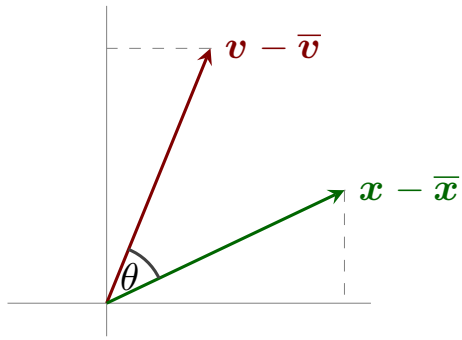
Estadística versus Mates II
(o la varianza vs el T^{ma} de Pitágoras)

F9

(**Lecture** 20) **S-10** **Covariance and correlation**

$$\sigma_{\boldsymbol{xy}} = \frac{1}{m}(\boldsymbol{x} - \boldsymbol{\mu_x}) \cdot (\boldsymbol{y} - \overline{\boldsymbol{y}});$$





$$\rho_{\boldsymbol{xy}} = \frac{\frac{1}{m}(\boldsymbol{x} - \boldsymbol{\mu_x}) \cdot (\boldsymbol{y} - \overline{\boldsymbol{y}})}{\|(\boldsymbol{x} - \boldsymbol{\mu_x})\| \cdot \|(\boldsymbol{y} - \overline{\boldsymbol{y}})\|} = \frac{\sigma_{\boldsymbol{xy}}}{\sqrt{\sigma_{\boldsymbol{x}}\sigma_{\boldsymbol{y}}}} = \cos(\theta).$$

F10

(**Lecture** 20) **S-11** **Ordinary Least Squares (OLS)**

Let $\mathbf{X}$ such that $\mathcal{L}\big([\mathbf{1};]\big) \subset \mathcal{C}\,(\mathbf{X})$.

$\widehat{\boldsymbol{y}}$ is the orthogonal projection of $\boldsymbol{y} \in \mathbb{R}^m$ onto $\mathcal{C}\,(\mathbf{X})$

$$\boxed{\widehat{\boldsymbol{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}} \quad \text{and} \quad \boxed{(\boldsymbol{y} - \widehat{\boldsymbol{y}}) \perp \mathcal{C}\,(\mathbf{X}) \;\Rightarrow\; \tfrac{1}{m}\mathbf{X}^\intercal(\boldsymbol{y} - \widehat{\boldsymbol{y}}) = \mathbf{0}}$$

$$\frac{1}{m}\mathbf{X}^\intercal(\boldsymbol{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}) = \mathbf{0} \iff \frac{1}{m}\mathbf{X}^\intercal\boldsymbol{y} - \frac{1}{m}\mathbf{X}^\intercal\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{0}.$$

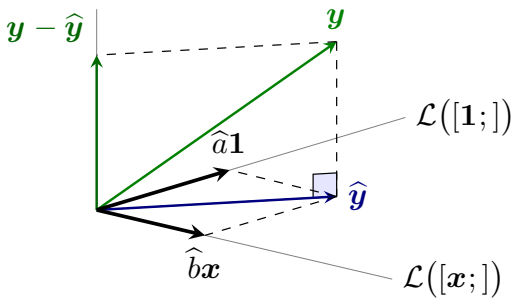
Therefore

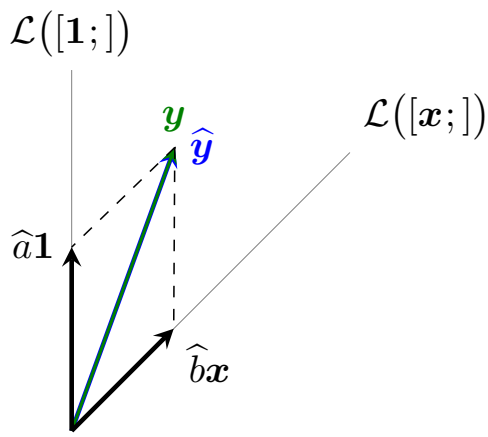
$$\Big(\frac{1}{m}\mathbf{X}^\intercal\mathbf{X}\Big)\widehat{\boldsymbol{\beta}} = \frac{1}{m}\mathbf{X}^\intercal\boldsymbol{y}.$$

F11

(**Lecture** 20) **S-12** **Ordinary Least Squares (OLS)**

If $\mathbf{X} = [\mathbf{1}; \boldsymbol{x};]$ has rank 2.



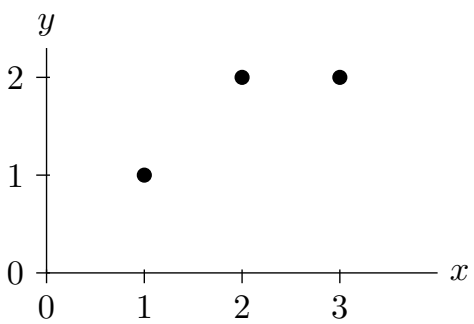


$$\left(\frac{1}{m}\mathbf{X}^\intercal\mathbf{X}\right)\begin{pmatrix}\widehat{a}\\ \widehat{b}\end{pmatrix} = \frac{1}{m}\mathbf{X}^\intercal\boldsymbol{y}.$$

F12

(**Lecture** 20) **S-13** **Application: Least Squares (Fitting by a line)**

"looking for the best-fitting line $\widehat{y} = \widehat{a} + \widehat{b}x$"
Points $(x, y,)$: $(1, 1,)$; $(2, 2,)$; $(3, 2,)$



$$\begin{cases} a + 1b &= 1 \\ a + 2b &= 2 \\ a + 3b &= 2 \end{cases} \quad \rightarrow \quad \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \qquad (\mathbf{X}\boldsymbol{\beta} = \boldsymbol{y} \text{ No solution})$$

F13

(**Lecture** 20) S-14 **Application: Least Squares (Fitting by a line)**

$$\mathbf{X}\boldsymbol{\beta} = \boldsymbol{y} \quad \text{(No solution)} \rightarrow \left(\frac{1}{m}\mathbf{X}^{\intercal}\mathbf{X}\right)\widehat{\boldsymbol{\beta}} = \frac{1}{m}\mathbf{X}^{\intercal}\boldsymbol{y} \quad \rightarrow \quad \widehat{\boldsymbol{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}.$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{pmatrix} \widehat{a} \\ \widehat{b} \end{pmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{pmatrix} \widehat{a} \\ \widehat{b} \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix} \quad \Rightarrow \quad \widehat{a} = \frac{2}{3}; \quad \widehat{b} = \frac{1}{2}.$$

$$\boxed{\text{Best solution: } \tfrac{2}{3} + \tfrac{1}{2}x}$$

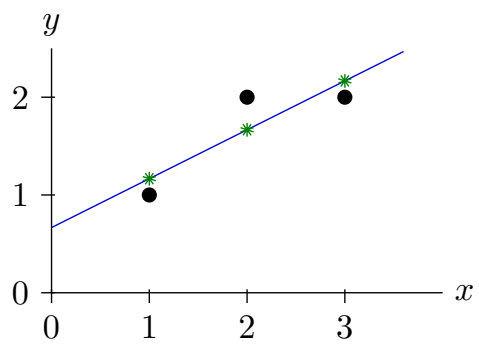
F14

$$\widehat{\boldsymbol{y}} = \mathbf{X}\widehat{\boldsymbol{b}} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{pmatrix} 2/3 \\ 1/2 \end{pmatrix} = \begin{pmatrix} 7/6 \\ 10/6 \\ 13/6 \end{pmatrix}$$

and

$$\widehat{\boldsymbol{e}} = \boldsymbol{y} - \widehat{\boldsymbol{y}} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 7/6 \\ 10/6 \\ 13/6 \end{pmatrix} = \begin{pmatrix} -1/6 \\ 2/6 \\ -1/6 \end{pmatrix}$$

(**Lecture** 20) S-15 **Application: Least Squares (Fitting by a line)**



$$\widehat{\boldsymbol{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{pmatrix} \widehat{a} \\ \widehat{b} \end{pmatrix} \longrightarrow \widehat{\boldsymbol{y}} = \begin{pmatrix} 7/6 \\ 10/6 \\ 13/6 \end{pmatrix} \longrightarrow \widehat{\boldsymbol{e}} = \begin{pmatrix} -1/6 \\ 2/6 \\ -1/6 \end{pmatrix}$$

$$\boldsymbol{y} = \widehat{\boldsymbol{y}} + \widehat{\boldsymbol{e}} \quad \text{and} \quad \begin{cases} \widehat{\boldsymbol{e}} \cdot \widehat{\boldsymbol{y}} & = 0 \\ \widehat{\boldsymbol{e}}\mathbf{X} & = \mathbf{0} \end{cases}.$$

F15

$$\widehat{\boldsymbol{e}} \cdot \widehat{\boldsymbol{p}} = \begin{pmatrix} -1/6, & 2/6, & -1/6, \end{pmatrix} \cdot \begin{pmatrix} 7/6 \\ 10/6 \\ 13/6 \end{pmatrix} = 0; \qquad \widehat{\boldsymbol{e}}\mathbf{A} = \begin{pmatrix} -1/6, & 2/6, & -1/6, \end{pmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{pmatrix} 0, & 0, \end{pmatrix}$$

Example 1. [**Selling price of homes:**]

Consider the following data. Selling prices (thousands of dollars) and living area (square feet) of 14 single-family homes in the *University City* community of San Diego, California, in 1990.

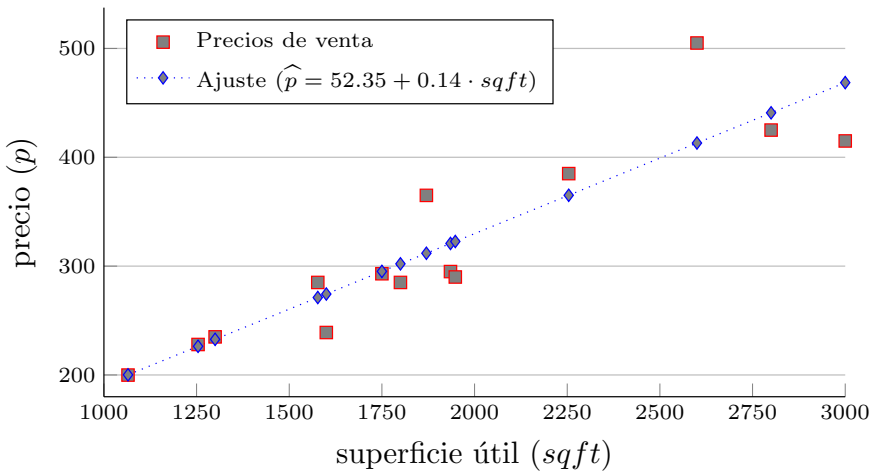
price = Sale price is in thousands of dollars

n	Price ($\boldsymbol{y}$)	Area ($\boldsymbol{x}$)
1	199.9	1065
2	228.0	1254
3	235.0	1300
4	285.0	1577
5	239.0	1600
6	293.0	1750
7	285.0	1800
8	365.0	1870
9	295.0	1935
10	290.0	1948
11	385.0	2254
12	505.0	2600
13	425.0	2800
14	415.0	3000

sqft = Square feet of living area(Ramanathan, 2002, pp. 78)

(Lecture 20) **S-16** **Application: Least Squares (Fitting by a line)**

Selling prices and living area (square feet) of 14 single-family homes in the *University City* community of San Diego, California, in 1990.
price = Sale price is in thousands of dollars
sqft = Square feet of living area (Ramanathan, 2002, pp. 78)



F16

Selling prices (thousands of dollars) and living area (square feet) of 14 single-family homes in the *University City*

n	Price	Living area	Fitted price	Error $\widehat{e}$
1	199.9	1065	200.1200	-0.22000
2	228.0	1254	226.3438	1.65619
3	235.0	1300	232.7263	2.27368
4	285.0	1577	271.1602	13.83984
5	239.0	1600	274.3514	-35.35142
6	293.0	1750	295.1640	-2.16397
7	285.0	1800	302.1015	-17.10148
8	365.0	1870	311.8140	53.18600
9	295.0	1935	320.8328	-25.83278
10	290.0	1948	322.6365	-32.63653
11	385.0	2254	365.0941	19.90587
12	505.0	2600	413.1017	91.89826
13	425.0	2800	440.8518	-15.85180
14	415.0	3000	468.6019	-53.60187

community of San Diego, California, in 1990.

(Ramanathan, 2002, pp. 78)

The lecture ends here

Questions of the Lecture 20

(L-20) Question 1. With the measurements $\boldsymbol{y} = (0, 8, 8, 20,)$ at $\boldsymbol{x} = (0, 1, 3, 4,)$,

(a) Set up and solve the normal equations $\mathbf{A}^\intercal\mathbf{A}\widehat{\boldsymbol{\beta}} = \mathbf{A}^\intercal\boldsymbol{y}$.
(b) For the best linear fit, find the four fitted values p_i and the four errors e_i.
(c) What is the square of the norm of the error vector $\|\boldsymbol{e}\|^2 = e_1^2 + e_2^2 + e_3^2 + e_4^2$?
(d) Draw the regression line
(e) Replace the measurements $\boldsymbol{y}$ with the fitted values $\boldsymbol{p} = (1, 5, 13, 17)$ write down the four equations $\mathbf{A}\boldsymbol{\beta} = \boldsymbol{p}$. Find an exact solution to $\mathbf{A}\boldsymbol{\beta} = \boldsymbol{p}$
(f) Verify that $\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{p} = (-1, 3, -5, 3)$ is perpendicular to both columns of the same matrix $\mathbf{A}$.
(g) What is the shortest distance $\|\boldsymbol{e}\|$ from $\boldsymbol{y}$ to the column space of $\mathbf{A}$?

(Strang, 2003, exercise 1–3 from section 4.3.)

(L-20) Question 2.

(a) Write down the three equations $y = \alpha + \beta x$ given the dataset: $y = 7$ at $x = -1$, $y = 7$ at $x = 1$, and $y = 21$ at $x = 2$. Find the least squares solution $\widehat{\boldsymbol{\beta}} = (\hat{\alpha}, \hat{\beta})$ and draw the best linear fit.
(b) Find the projection $\boldsymbol{p} = \mathbf{A}\widehat{\boldsymbol{\beta}}$. That is, three values of the best linear fit. Show that the error vector is $\boldsymbol{e} = (2, -6, 4)$. Why is $\mathbf{P}\boldsymbol{e} = \mathbf{0}$?

(L-20) Question 3. Our measurements at times $t = 1, 2, 3$ are $b = 1, 4$, and b_3. We want to fit those points by the nearest line $C + Dt$, using least squares.

(a) Which value for b_3 will put the three measurements on a straight line? *Which line is it?* Will least squares choose that line if the third measurement is $b_3 = 9$? (Yes or no).
(b) What is the linear system $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ that would be solved exactly for $\boldsymbol{x} = (C, D)$ if the three points do lie on a line? Compute the projection matrix $\mathbf{P}$ onto the column space of $\mathbf{A}$.
(c) What is the rank of that projection matrix $\mathbf{P}$? How is the column space of $\mathbf{P}$ related to the column space of $\mathbf{A}$? (You can answer with or without the entries of $\mathbf{P}$ computed in (b).)
(d) Suppose $b_3 = 1$. Write down the equation for the best least squares solution $\widehat{\boldsymbol{x}}$, and show that the best straight line is horizontal.

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(L-20) Question 4. Graphically represent the mean and variance of the vector $\boldsymbol{x} = (2, 4)$ on a grid.

End of Questions of the Lecture 20

References

Ramanathan, R. (2002). *Introductory Econometrics with applications*. South-Western, Mason, Ohio, fifth ed. ISBN 0-03-034186-8.

Strang, G. (2003). *Introduction to Linear Algebra*. Wellesley-Cambridge Press, Wellesley, Massachusetts. USA, third ed. ISBN 0-9614088-9-8.

Solutions

(L-20) Question 1(a) For the vectors $\boldsymbol{y}$ and $\boldsymbol{x}$, the matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \quad \boldsymbol{y} = \begin{pmatrix} 0 \\ 8 \\ 8 \\ 20 \end{pmatrix}$$

The normal equations are given by $\mathbf{A}^\intercal \mathbf{A} \widehat{\boldsymbol{\beta}} = \mathbf{A}^\intercal \boldsymbol{y}$, or

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{pmatrix} \widehat{\alpha} \\ \widehat{\beta} \end{pmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \end{bmatrix} \begin{pmatrix} 0 \\ 8 \\ 8 \\ 20 \end{pmatrix}$$

or

$$\begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \begin{pmatrix} \widehat{\alpha} \\ \widehat{\beta} \end{pmatrix} = \begin{pmatrix} 36 \\ 112 \end{pmatrix}$$

which has as its solution $\widehat{\alpha} = 1$, $\widehat{\beta} = 4$.

□

(L-20) Question 1(b) Thus, the four fitted values with this $\widehat{\boldsymbol{\beta}}$ are given by

$$\boldsymbol{p} = \mathbf{A}\widehat{\boldsymbol{\beta}} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 13 \\ 17 \end{pmatrix}.$$

With this solution, the error vector $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ is given by

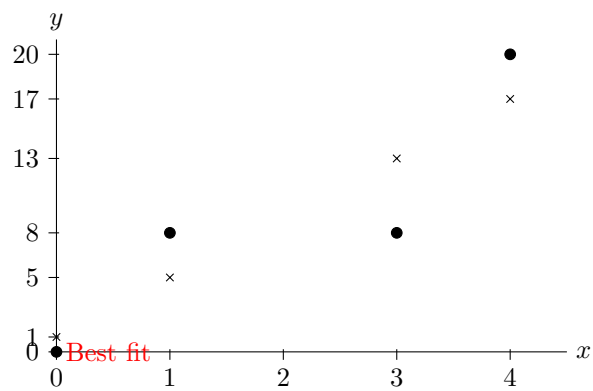
$$\boldsymbol{e} = \begin{pmatrix} 0 \\ 8 \\ 8 \\ 20 \end{pmatrix} - \begin{pmatrix} 1 \\ 5 \\ 13 \\ 17 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ -5 \\ 3 \end{pmatrix}.$$

□

(L-20) Question 1(c) It is the smallest possible value for a linear fit: $\|\boldsymbol{e}\|^2 = \boldsymbol{e} \cdot \boldsymbol{e} = (-1)^2 + 3^2 + (-5)^2 + 3^2 = 44$.

□

(L-20) Question 1(d)



□

(L-20) Question 1(e) If the new values change to those given in the statement, then we have

$$\begin{aligned}\alpha + 0\beta =& 1\\ \alpha + 1\beta =& 5\\ \alpha + 3\beta =& 13\\ \alpha + 4\beta =& 17\end{aligned}$$

Which has an analytic solution given by $\alpha = 1$ and $\beta = 4$.

□

(L-20) Question 1(f)

$$\boldsymbol{e}\mathbf{A} = \begin{pmatrix}-1, & 3, & -5, & 3,\end{pmatrix}\begin{bmatrix}1 & 0\\ 1 & 1\\ 1 & 3\\ 1 & 4\end{bmatrix} = \begin{pmatrix}0, & 0,\end{pmatrix}.$$

□

(L-20) Question 1(g) The shortest distance is given by $\|\boldsymbol{e}\| = \sqrt{\boldsymbol{e}\cdot\boldsymbol{e}} = \sqrt{44}$.

□

(L-20) Question 2(a) Our equations are given by

$$\begin{aligned}\alpha - \beta =& 7\\ \alpha + \beta =& 7\\ \alpha + 2\beta =& 21\end{aligned}$$

Which can be written as a system of linear equations

$$\begin{bmatrix}1 & -1\\ 1 & 1\\ 1 & 2\end{bmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \begin{pmatrix}7\\ 7\\ 21\end{pmatrix}$$

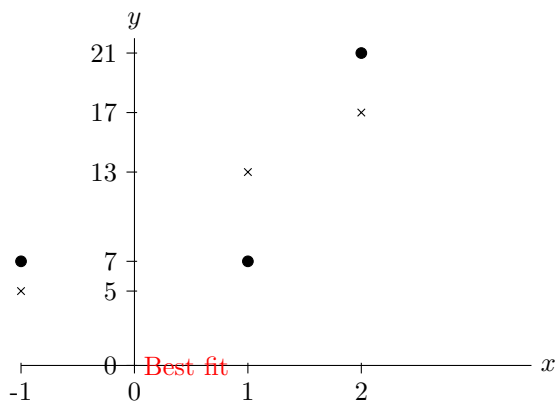
The least squares solution is given by $\mathbf{A}^\intercal\mathbf{A}\widehat{\boldsymbol{\beta}} = \mathbf{A}^\intercal\boldsymbol{y}$ which in this case simplifies as follows

$$\begin{bmatrix}1 & 1 & 1\\ -1 & 1 & 2\end{bmatrix}\begin{bmatrix}1 & -1\\ 1 & 1\\ 1 & 2\end{bmatrix}\begin{pmatrix}\widehat{\alpha}\\ \widehat{\beta}\end{pmatrix} = \begin{bmatrix}1 & 1 & 1\\ -1 & 1 & 2\end{bmatrix}\begin{pmatrix}7\\ 7\\ 21\end{pmatrix}$$

or

$$\begin{bmatrix}3 & 2\\ 2 & 6\end{bmatrix}\begin{pmatrix}\widehat{\alpha}\\ \widehat{\beta}\end{pmatrix} = \begin{pmatrix}35\\ 42\end{pmatrix}$$

which gives the following values $\widehat{\alpha} = 9$, $\widehat{\beta} = 4$. Thus, the best linear fit is $\widehat{y} = 9 + 4x$.



□

(L-20) Question 2(b)

$$\boldsymbol{p} = \mathbf{A}\widehat{\boldsymbol{\beta}} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{pmatrix} 9 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ 13 \\ 17 \end{pmatrix}.$$

which gives the values of the best linear fit. The error vector $\boldsymbol{e}$ is then given by

$$\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{p} = \begin{pmatrix} 7 \\ 7 \\ 21 \end{pmatrix} - \begin{pmatrix} 5 \\ 13 \\ 17 \end{pmatrix} = \begin{pmatrix} 2 \\ -6 \\ 4 \end{pmatrix}.$$

The matrix $\mathbf{P}$ projects onto the column space of $\mathbf{A}$, and $\boldsymbol{e}$ lies in the left null space, thus it is orthogonal to $\mathcal{C}\left(\mathbf{A}\right)$.

□

(L-20) Question 3(a) The three data points lie on the same line when $b_3 = 7$. This line is $-2 + 3t$. If $b_3 = 9$, the least squares method will NOT choose this line. (A quick way to see this is from the fact that the line chosen by least squares will give the average of the given b's at the time equal to the average of the given t's; in this case, the best fit line would take the value $(1 + 3 + 9)/3 = 13/3$ at $t = (1 + 2 + 3)/3 = 2$, whereas our line gives 4 at $t = 2$.)

□

(L-20) Question 3(b) The linear system for $\boldsymbol{x} = (C, D)$ would be the following:

$$\mathbf{A}\boldsymbol{x} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ b_3 \end{pmatrix}.$$

We compute the projection matrix $\mathbf{P}$ onto the column space of $\mathbf{A}$ using the projection matrix formula:

$$\begin{aligned} \mathbf{P} =& \mathbf{A}(\mathbf{A}^\intercal\mathbf{A})^{-1}\mathbf{A}^\intercal = \frac{1}{6}\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \\ =& \frac{1}{6}\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 8 & 4 & -4 \\ -3 & 0 & 3 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}. \end{aligned}$$

□

(L-20) Question 3(c) The column space of $\mathbf{P}$ is the space consisting of all the vectors $\mathbf{P}\boldsymbol{b}$, i.e., all the projections of vectors in $\mathbb{R}^3$ onto the column space of $\mathbf{A}$, which is precisely the column space of $\mathbf{A}$. Thus, the rank of $\mathbf{P}$ is equal to the rank of $\mathbf{A}$, which is 2.

□

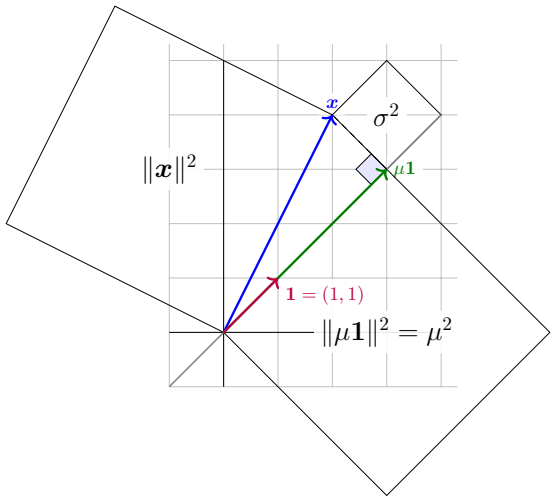
(L-20) Question 3(d) The equation for the best least squares solution $\widehat{\boldsymbol{x}}$ is $\mathbf{A}^\intercal\mathbf{A}\widehat{\boldsymbol{x}} = \mathbf{A}^\intercal\boldsymbol{b}$, where $\boldsymbol{b} = \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}$. Writing out this system, we get

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \widehat{\boldsymbol{x}} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 12 \end{pmatrix}.$$

The solution to this system is $\widehat{\boldsymbol{x}} = \begin{pmatrix} C \\ D \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, so the best fit line is the horizontal line $b = 2$.

□

(L-20) Question 4.


$\boldsymbol{x}$
σ^2
$\|\boldsymbol{x}\|^2$
$\mu\mathbf{1}$
$\mathbf{1} = (1, 1)$
$\|\mu\mathbf{1}\|^2 = \mu^2$

$\square$